PsiSAT

Physics Proposal: Path A

A 1U Educational CubeSat Counts/Timing Pathfinder
for Gamma-Ray Burst Detection in Low Earth Orbit

Stony Brook University

Target program

NASA CubeSat Launch Initiative (CSLI)

Goal: Build and fly a small student-built satellite (a 1U CubeSat) that carries a gamma-ray detector. The satellite will record the times at which the gamma-ray count rises above its normal level, share those times publicly, and show that the whole chain from detector to data on the ground works in orbit.

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1. Abstract

PsiSAT (also called LEO-GRM) is a small satellite proposed by students at Stony Brook University. It is a 1U CubeSat, which means a satellite shaped like a cube about 10 cm on each side. Its instrument is a flat crystal of cesium iodide doped with thallium, written CsI(Tl), that gives off a tiny flash of light whenever a gamma ray hits it. A light sensor called a silicon photomultiplier (SiPM) turns each flash into an electrical pulse. An amplifier makes the pulse larger, and a comparator (a circuit that switches on only when a pulse is bigger than a chosen level) turns it into a digital signal. A small computer chip, the STM32 microcontroller, records the time of each signal.

This proposal describes **Path A**, the first version of the mission. Path A only counts how many gamma rays arrive and when they arrive. It does *not* measure the energy of each gamma ray (a technique called pulse-height analysis, or PHA). Because of this, Path A cannot produce energy spectra, and it cannot be matched against gravitational-wave detections such as those from LIGO. Those abilities are left for later upgrades (Paths B and C, explained in Section 2).

The crystal measures $50 \times 50 \times 3$ mm, so its face has an area of $A_{\text{geom}} = 25$ cm^2. Using tabulated absorption data from NIST (the U.S. National Institute of Standards and Technology) and a CsI density of $\rho = 4.51$ g/cm^3 (*Assumed*), a gamma ray of 100 keV is absorbed in CsI with an attenuation coefficient of $\mu \approx 9.18$ cm^{-1} (*Derived*). The chance that a 100 keV gamma ray, arriving head-on at the crystal face, interacts with the crystal is $\alpha(100\,keV) = 1 - e^{-\mu x} \approx 93.6\%$ (*Derived*). This number describes the crystal only. It is *not* the fraction of bursts the satellite would detect.

The quantity that matters for sensitivity is the *effective area*, the area of a perfect detector that would count the same number of gamma rays:

$$A_{\text{eff}}(E) \approx A_{\text{geom}} \times \alpha(E) \times \varepsilon_{\text{optics+SiPM}}(E) \times \varepsilon_{\text{thr}}(E).$$

The last two factors (how well the light is collected and converted to a pulse, and how many pulses are large enough to be counted) have not been measured yet. They will be measured in the laboratory.

2. Terms and conventions used in this document

2.1 Path A, Path B, Path C

- **Path A** (this proposal): counts gamma rays and records when they arrive. No energy measurement and no GPS clock.
- **Path B** (possible later upgrade): adds pulse-height analysis, so the energy of each gamma ray is measured.
- **Path C** (possible later upgrade): adds a GPS receiver, so that timestamps are tied to an accurate universal clock.

2.2 How numbers are labeled

Every performance number in this document carries one of four labels:

- *Measured*: taken from a measurement or from a published table.
- *Derived*: calculated from other numbers.

- *Assumed*: a value we chose or took as typical, not yet checked.
- **TBD** (to be determined): not known yet. A plan to find it is given.

2.3 Glossary

Term	Meaning
GRB (gamma-ray burst)	A very bright flash of gamma rays from a distant explosion, lasting from a fraction of a second to a few minutes.
Gamma ray, keV, MeV	Gamma rays are high-energy light. Their energy is measured in keV (thousand electron-volts) or MeV (million electron-volts).
LEO (low Earth orbit)	An orbit a few hundred kilometers above Earth.
CubeSat, 1U, 3U, 6U	A standard small satellite built from 10 cm cubes. 1U is one cube, 3U is three cubes joined together, and so on.
Pathfinder	An early, small mission whose main purpose is to show that a technology or method works in space, so that later missions can rely on it.
CsI(Tl)	Cesium iodide with a little thallium added. A crystal that flashes visible light when a gamma ray or charged particle deposits energy in it (a *scintillator*).
SiPM (silicon photomultiplier)	A light sensor that turns very weak flashes of light into electrical pulses.
PDE	Photon detection efficiency: the fraction of light particles reaching a SiPM that produce a pulse.
Comparator	An electronic circuit that gives a digital signal only when the input pulse is larger than a set level (the *threshold*).
STM32	A family of small computer chips (microcontrollers) made by STMicroelectronics.
PHA (pulse-height analysis)	Measuring the height of each pulse, which is proportional to the energy of the gamma ray.
Effective area	The area of a perfect detector that would count the same number of gamma rays as the real detector.
Background	Counts that are not from the burst we are looking for: cosmic rays, charged particles, radiation from Earth, and electronic noise.
False alarm	A rise in counts that looks like a burst but is caused by background or noise.
SAA (South Atlantic Anomaly)	A region over the South Atlantic where Earth's magnetic field lets more charged particles reach low altitudes, so the count rate is much higher.
GPS	Global Positioning System. Also provides an accurate time signal.
Mission elapsed time	Time counted from the moment the satellite's clock started, rather than a calendar date and clock time.
LIGO, gravitational waves	LIGO is an observatory that detects ripples in space-time (gravitational waves), for example from merging neutron stars.

Term	Meaning
GCN	The Gamma-ray Coordinates Network (now called the General Coordinates Network), a public service through which observatories announce bursts and other events.
NIST XCOM	An online database from NIST giving how strongly each material absorbs X-rays and gamma rays at each energy.
TRL (technology readiness level)	A NASA scale from 1 to 9 that describes how mature a technology is. Low numbers mean an idea or lab test, and high numbers mean proven in real missions.
CSLI	NASA's CubeSat Launch Initiative, a program that selects small satellites for launch opportunities.

3. Science background: gamma-ray bursts

3.1 What gamma-ray bursts are

Gamma-ray bursts (GRBs) are the most luminous explosions known. They release most of their energy as hard X-rays and soft gamma rays, in flashes that last from milliseconds to minutes. Their brightness can change in just a few milliseconds [1, 2, 16]. The gamma rays typically have energies from tens of keV to a few MeV. Astronomers usually divide GRBs into two groups [1, 2]:

Long GRBs last longer than about 2 seconds. (The standard measure of duration is T_{90}, the time in which 90% of the burst's gamma rays arrive.) They usually have softer spectra, with most of their energy near 150 keV, and are mainly linked to the collapse of massive stars into black holes that launch fast jets of matter [1, 2].

Short GRBs last less than about 2 seconds. They are usually harder, with most of their energy near 450 keV, and are linked to the merger of two compact stars (two neutron stars, or a neutron star and a black hole). The short GRB 170817A was the first gamma-ray burst seen together with a gravitational-wave signal, from the neutron-star merger GW170817 [1, 3].

3.2 Why small, low-cost detectors are useful

GRBs are bright and short, so even small detectors can see them. The first GRBs were discovered with small CsI scintillators on the Vela satellites [1]. Today, astronomers who study bursts together with gravitational waves care more about watching the whole sky all the time and sending fast alerts than about very precise energy measurements [1]. Small satellites help in four ways:

- **More sky coverage and backup.** Several small satellites leave fewer gaps when Earth blocks the view, and the loss of one satellite does not end the observations [1].
- **Locating bursts.** If a burst reaches several satellites far apart at slightly different times, the time differences can be used to find where in the sky it came from. One satellite alone cannot do this well [1, 4, 5].
- **Testing technology.** Flying SiPM-and-scintillator detectors in space raises their technology readiness level (TRL), so that larger missions can use them with less risk. It also trains

students [1].
- **Education.** Students go through a full mission, from the detector to data received on the ground.

3.3 How many bursts do large instruments see?

The Fourth Fermi-GBM GRB Catalog covers 10 years (12 July 2008 to 11 July 2018) and lists 2356 cosmic GRBs, about 236 per year, seen by the large sodium-iodide and bismuth-germanate detectors of NASA's Fermi Gamma-ray Burst Monitor [6, 13] (*Measured* by Fermi-GBM; this is not a PsiSAT prediction). The earlier BATSE instrument on the Compton Gamma Ray Observatory saw about 300 per year with a wide field of view [6]. These rates cannot be applied to a 25 cm^2 detector like PsiSAT's. To estimate a PsiSAT rate, we first need its effective area, the fraction of time it can observe, its field of view and orientation, and the faintest burst it can detect. See Section 12.1. This estimate is currently **TBD**.

4. Goal and scope of Path A

4.1 Goal

The goal of Path A is to show that a student-built 1U CubeSat for the NASA CSLI program can do all of the following steps in orbit, in this order:

1. A gamma ray or charged particle deposits energy in the CsI(Tl) crystal.
2. The crystal gives off a flash of light.
3. The SiPM turns the flash into an electrical pulse.
4. The amplifier enlarges the pulse.
5. The comparator produces a digital signal if the pulse is large enough.
6. The STM32 records the time of the signal.
7. Software checks whether the count rate has risen above normal.
8. The satellite stores a record of each rise.
9. The records are sent to the ground.

The times of these rises will be published openly, as mission elapsed time. The team will also document how the background radiation, the South Atlantic Anomaly (SAA), and the SiPM's noise behave in orbit, in the same way that the GRBAlpha team did for its satellite [1, 4, 5].

4.2 What Path A will do

Activity	What it means
Count gamma-ray-like events and record their arrival times	Each detected event becomes one digital signal, and the microcontroller stores the time it arrived.
Report when the count rate rises above normal	The science result is a list of times when the rate was unusually high, not an energy spectrum.
Publish these times openly	Anyone can compare them with bursts reported by other observatories.
Show that students can build and run the whole chain	From detector to data on the ground, including testing and operations.
Compare fairly with similar missions	See Section 5. [1, 4, 5]
Record background, SAA, and SiPM aging in orbit	Any detector of this type in low Earth orbit must deal with these effects [1, 4].
Describe possible upgrades only as future work	Path B (energy measurement) and Path C (GPS time) are not part of this mission.

4.3 What Path A does not claim

Not claimed	Reason
An overall detection efficiency	The efficiency has not been measured yet (Section 7).
A percentage of all GRBs detected	The interaction probability α only tells how likely a gamma ray that hits the crystal is to interact with it. It says nothing about how many bursts we would see.
The faintest burst we can detect (fluence limit)	This needs the effective area, the threshold, and a background model.
Energy spectra, or telling soft bursts from hard ones	Path A does not measure the energy of each gamma ray.
Coincidence with LIGO or other gravitational-wave detections	There is no GPS clock, so timestamps are not tied to universal time.
A magnet that points the detector away from Earth	A permanent magnet lines the satellite up with Earth's magnetic field, not with the direction to Earth's center.
The number of GRBs detected per year	This needs the exposure, the fraction of time observing, and the threshold.
Watching the whole sky or locating bursts	There is one flat detector and the satellite's orientation is not controlled.

4.4 Summary of the instrument

Signal chain: gamma ray or particle → light in the CsI crystal → SiPM → amplifier → comparator (threshold) → digital signal → time recorded by the STM32.

Orientation assumption: the satellite is not actively pointed. It is assumed to tumble or settle in a random orientation, possibly slowed by a magnet. We do not assume it faces away from Earth.

Timing: mission elapsed time only, until GPS is added in Path C.

Detector: CsI(Tl) crystal $50 \times 50 \times 3$ mm, read out by a Hamamatsu S13360-6050PE-class SiPM (from the current parts list). The face area facing the source is $A_{\text{geom}} = 25$ cm^2.

5. Similar CubeSat and small-satellite missions

This section summarizes similar missions, using the review by Bloser et al. (2022) [1] and the primary papers. PsiSAT uses the same kind of detector as GRBAlpha (a CsI crystal read out by SiPMs on a 1U satellite), but it is a simpler instrument: it does not measure energy in Path A, it has a smaller and thinner crystal, its effective area has not been published, and it has not flown.

5.1 GRBAlpha (the closest match)

GRBAlpha is a 1U CubeSat that has already flown. It was launched on 22 March 2021 into a polar orbit about 550 km high that keeps a fixed orientation to the Sun (a sun-synchronous orbit) [4, 5]. Its detector is a $75 \times 75 \times 5$ mm CsI(Tl) crystal wrapped in a reflective film (ESR) and read out by eight Hamamatsu S13360-3050PE light sensors (MPPCs, a type of SiPM) arranged in a 2×4 grid. The design also has a light-tight wrap, an anodized aluminum housing, and lead-alloy shielding at the edge of the sensors [4]. Its published energy range is about 70–890 keV with an energy resolution of about 30%. At launch the lowest detectable energy (the threshold) was near 30 keV. After about two years, radiation damage to the SiPMs had raised it to about 60–70 keV [4].

The crystal face of GRBAlpha is 56.25 cm^2, about 2.25 times PsiSAT's, and its crystal is 5 mm thick, compared with PsiSAT's 3 mm. GRBAlpha also records the energy of the gamma rays, in binned spectra. PsiSAT Path A will not. Maps of its count rate showed higher counts over the poles and over the SAA. The fraction of time in which it could detect an average GRB at the 5-sigma level was quoted as about 67% for that orbit [4] (*Measured* by GRBAlpha). At first, it recorded count rates in continuous time bins, and an automatic onboard trigger was a planned upgrade [4]. It has detected several confirmed GRBs, including GRB 221009A, which it recorded without its detector saturating [4, 5]. GRBAlpha is a pathfinder for the CAMELOT constellation [1, 5, 14].

Lessons for PsiSAT: (1) A 1U satellite with a CsI crystal and SiPMs can detect real GRBs, as long as the claims match what the hardware can do. (2) The effective area versus energy and angle should be published, from simulation or flight data. (3) SiPM noise should be expected to rise over time. (4) The SAA and the polar regions are the main cause of false alarms. (5) A single 1U satellite is a pathfinder, not a burst locator.

5.2 VZLUSAT-2

VZLUSAT-2 is a 3U Czech technology CubeSat launched in January 2022 into a polar orbit about 535 km high. It carries two detectors of the same size as GRBAlpha's ($75 \times 75 \times 5$ mm CsI with SiPMs), mounted perpendicular to each other [1, 5]. The quoted energy range is about 30–890 keV [1]. It has detected GRBs and solar flares. It has active attitude control, but the time available for the gamma-ray instrument is limited by power and data budgets (about one third of the time in later reports) [5]. It operated successfully for multiple years [1, 5].

Lesson: two detectors facing different directions help, and power and data limits reduce the observing time.

5.3 BurstCube (NASA Goddard)

BurstCube is a 6U CubeSat built to detect and locate GRBs, especially short GRBs that come with gravitational waves [1, 7, 8]. It has four CsI(Tl) disks, each 90 mm across and 19 mm thick, and each disk is read out by 116 Hamamatsu S13360-6050 SiPMs. The disks are tilted about 45° away from straight up (the zenith) [7, 8]. Its energy range is about 50 keV–1 MeV. Simulations in the BurstCube papers give one crystal an effective area at 100 keV of about 70% of one Fermi-GBM sodium-iodide detector [7, 8] (*Derived*, from computer simulations with the MEGAlib and Geant4 software). It has an onboard trigger and sends rapid alerts through NASA's relay satellites (TDRSS, the Space Network). It was deployed from the International Space Station in April 2024, and its science operations were shorter than planned, ending in September 2024 according to the HEASARC mission page [8]. Before launch, the team expected about 20 short and 10 long GRBs per year [8]. These numbers do not apply to PsiSAT.

Lesson: locating bursts with several detectors and sending alerts with accurate absolute time is a much more advanced capability than Path A. PsiSAT should not use BurstCube's or LIGO's language unless Path B and Path C hardware exist.

5.4 HERMES Pathfinder

HERMES is a planned group of several 3U nanosatellites. Each uses GAGG scintillator crystals read out by silicon drift detectors (SDDs). The aim is to locate bursts from small differences in arrival time, with timing goals below one microsecond. The quoted energy range is about 5–500 keV [1, 15]. SpIRIT, an Australian satellite, carries a HERMES-type instrument [1].

Lesson: locating bursts from arrival times needs several satellites and very accurate absolute time. This is Path C territory.

5.5 GRID (Chinese student CubeSats)

GRID is a student-led project. Each detector has a 2×2 array of GAGG crystals (each $3.8 \times 3.8 \times 1$ cm^3; GAGG is a different scintillator material) read out by SiPM arrays from SensL/On Semiconductor, covering about 10 keV–2 MeV [1, 9]. The first units flew between 2018 and 2022 on 6U satellites in polar low Earth orbit, and several GRBs were detected. GRID-02 showed the SiPM dark current (the electrical current present even without light) rising by about 100 μA per year per chip [1]. More than 150 students from about 20 institutions took part [1].

Lesson: student-built missions of this kind are realistic, and SiPM aging in orbit is a major result that every such mission should report.

5.6 SIRI and Glowbug (U.S. Naval Research Laboratory)

SIRI and SIRI-2 flew a SiPM with a strontium-iodide scintillator (SrI_2) in orbit, through the U.S. Department of Defense Space Test Program. Their successors include GARI (with GAGG) and Glowbug, a concept for a gamma-ray transient telescope covering about 30 keV–2 MeV [1, 10]. These missions improve the readiness and radiation-damage knowledge of SiPMs for larger missions.

Lesson: these are useful sources on SiPM technology, but they are not comparable in performance to PsiSAT.

5.7 CAMELOT (planned constellation)

CAMELOT is a proposed fleet of 3U CubeSats. Each would carry large, thin CsI(Tl) crystals (about $150 \times 75 \times 5$ mm) with SiPMs on perpendicular walls, to monitor GRBs and to locate them from arrival-time differences [1, 5, 14, 17]. GRBAlpha and VZLUSAT-2 are its pathfinders. In the sources we consulted, the full constellation had not yet flown.

Lesson: PsiSAT is a pathfinder for education and technology. It is not a burst-locating constellation.

5.8 EIRSAT-1 and its GMOD instrument

EIRSAT-1 is Ireland's 2U educational CubeSat, part of the European Space Agency's "Fly Your Satellite!" program. It carries GMOD, which has a $25 \times 25 \times 40$ mm cerium-bromide ($CeBr_3$) scintillator read out by 4×4 SensL J-series SiPMs and a dedicated chip, SIPHRA [1, 11]. Its energy range is about 30 keV–2 MeV. Its energy resolution is 5.4% at 662 keV (*Measured* on the engineering model) [11]. Its sky-averaged effective area is about 10 cm^2, from computer simulations using MEGAlib [11] (*Derived*). The team estimated 11 to 14 GRBs per year detected above 10 sigma for a one-year mission [11]. This estimate depends on assumptions, and GMOD measures energy, unlike PsiSAT Path A.

Lesson: even a small educational detector publishes its simulated effective area and energy resolution. PsiSAT Path A must state that its effective area is **TBD** until laboratory measurements and simulations fix the efficiency factors.

5.9 Comparison table

Mission	Size	Detector	Energy range (published)	Measures energy?	Status
PsiSAT	1U (planned)	CsI(Tl) $50 \times 50 \times 3$ mm + SiPM → amplifier → comparator	**TBD**	No, counts only in Path A	Student project, not yet built or flown
GRBAlpha	1U, flown	CsI(Tl) $75 \times 75 \times 5$ mm + 8 MPPCs	∼70–890 keV (∼30% resolution)	Yes	Launched 2021; detected several GRBs [4]
VZLUSAT-2	3U, flown	Two GRBAlpha-type CsI + SiPM detectors	∼30–890 keV	Yes	Launched 2022; detected GRBs and solar flares [1, 5]
BurstCube	6U, flown	4 CsI disks, 90 mm × 19 mm, 116 SiPMs each	∼50 keV–1 MeV	Yes	Deployed from ISS in 2024; aimed at gravitational-wave counterparts [7, 8]
CAMELOT	3U fleet (planned)	CsI $150 \times 75 \times 5$ mm + SiPM	GRB monitoring	Yes (planned)	Full fleet not yet flown [1]
GRID	6U units	2×2 GAGG + SiPM	∼10 keV–2 MeV	Yes	Several launches 2018–2022 [1, 9]
HERMES Pathfinder	3U fleet	GAGG + silicon drift detector	∼5–500 keV	Yes	Demonstrating burst location by timing [1]
EIRSAT-1 GMOD	2U	$CeBr_3$ $25 \times 25 \times 40$ mm + 16 SiPMs	∼30 keV–2 MeV	Yes (5.4% at 662 keV)	Educational; effective area ∼10 cm^2 from simulation [11]
SIRI / Glowbug	Various	SrI_2 or GAGG + SiPM	Depends on mission	Yes	Technology demonstrations of SiPMs [1, 10]

Table 2: Comparison with similar missions. PsiSAT Path A uses the same kind of detector as GRBAlpha but is simpler, since it only counts pulses. Its effective area is **TBD**.

5.10 Where PsiSAT stands

Compared with GRBAlpha, PsiSAT has a smaller and thinner crystal, does not measure energy in Path A, and has no flight calibration. It is one or two major steps behind. Compared with BurstCube, it is a much simpler kind of mission. PsiSAT is best described as an educational counts-and-timing demonstrator inspired by GRBAlpha.

6. How the detector works

6.1 The signal chain in eight steps

1. **Arrival.** A gamma ray reaches the crystal face. Whether it does depends on the satellite's orientation, its field of view, and whether Earth blocks the source (occultation).
2. **Interaction.** With probability $\alpha(E) = 1 - e^{-\mu(E)x}$, the gamma ray interacts with the crystal (by the photoelectric effect, Compton scattering, or other processes). This probability is not the percentage of GRBs detected.
3. **Energy deposit.** Part or all of the gamma ray's energy stays in the crystal.
4. **Scintillation.** Some of the deposited energy becomes flashes of visible light. CsI(Tl) produces

about 54 light particles per keV, at a wavelength near 550 nm. This light yield describes the crystal and is not a detection efficiency.

5. **Light transport.** The wrap, window, and optical gel determine the fraction of the light that reaches the SiPM, f_{coll} (**TBD**).
6. **SiPM.** The SiPM converts the light into an electrical pulse. The size of the pulse depends on the photon detection efficiency (PDE), the gain, the temperature, and radiation damage. The SiPM also produces *dark pulses*, which are random pulses that occur without any light.
7. **Electronics.** The amplifier shapes the pulse. The comparator accepts the pulse only if it is larger than the set threshold.
8. **Counting.** Each accepted pulse triggers an input on the STM32 (an external interrupt, called EXTI), and the STM32 records the time. The counts are added up in time bins and compared with the expected background. A count above the expected level raises a flag.

6.2 Interaction probability (the Beer–Lambert law)

Consider a narrow beam of gamma rays, all with the same energy E, hitting a slab of thickness x head-on. The fraction that passes through without interacting is

$$I/I_0 = e^{-\mu(E)\,x},$$

so the fraction that interacts at least once is

$$\alpha(E) = 1 - e^{-\mu(E)\,x}.$$

Here $\mu(E)$ is the linear attenuation coefficient, which tells how strongly the material absorbs gamma rays of energy E.

$\alpha(E)$ **is** the fraction of gamma rays that hit the crystal face and interact at least once at energy E.

$\alpha(E)$ **is not** the fraction of all GRBs detected, an efficiency averaged over a range of energies, the overall detection efficiency of the mission, or proof that a digital count was recorded.

6.3 Checking the number at 100 keV with NIST data

The NIST XCOM database for cesium iodide gives a mass attenuation coefficient $\mu/\rho(100\,keV) = 2.035$ cm^2/g [12] (*Measured*, from a published table). With $\rho = 4.51$ g/cm^3 (*Assumed*, a typical value for crystalline CsI), and a thickness $x = 0.3$ cm:

$$\mu = (\mu/\rho)\cdot\rho = 2.035 \times 4.51 \approx 9.18 \text{ cm}^{-1} \quad (\textit{Derived})$$
$$\alpha(100\,keV) = 1 - e^{-9.18\times 0.3} \approx 0.936\ (93.6\%) \quad (\textit{Derived})$$

The total attenuation coefficient slightly overstates how much energy is fully absorbed, because some scattered gamma rays escape the crystal. In Path A we therefore call α an interaction probability. A later analysis can use the energy-absorption coefficient μ_{en} or a Geant4 simulation (a standard particle-physics simulation program).

6.4 The interaction probability from 20 keV to 1 MeV

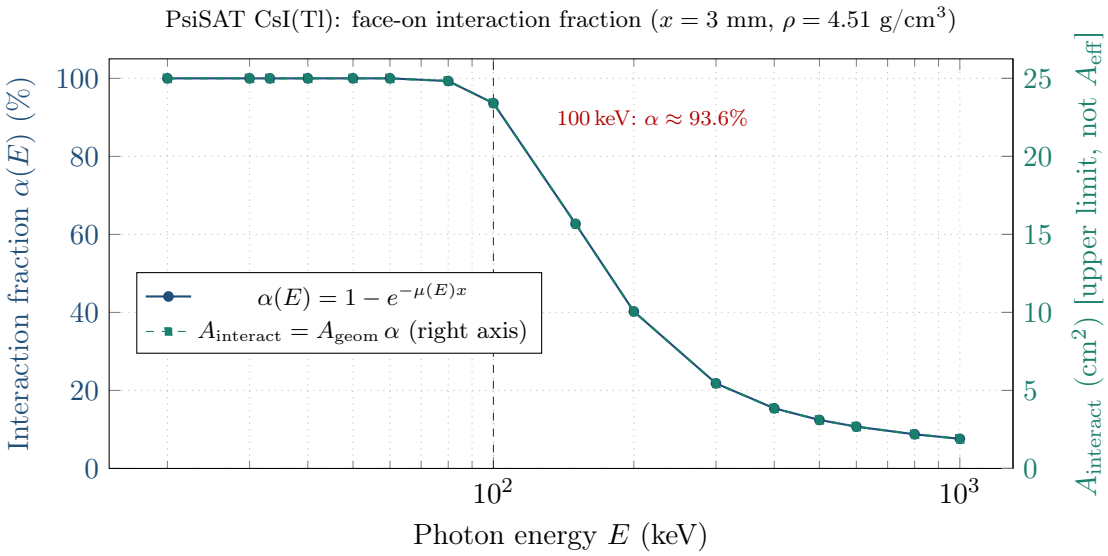


Figure 1: Interaction fraction $\alpha(E)$ (left axis) and $A_{\text{interact}}(E) = A_{\text{geom}} \cdot \alpha(E)$ (right axis) for a 3 mm CsI crystal viewed head-on, from NIST XCOM data. A_{interact} is an upper limit on the effective area, not the effective area itself.

Almost every gamma ray below about 80 keV interacts in the 3 mm crystal. At 100 keV about 94% interact, at 200 keV about 40%, and at 300 keV about 22% (*Derived*). A single percentage for the whole energy range would therefore be misleading. The full table is in Appendix A.

7. Instrument response and effective area

7.1 Definitions

Symbol	Meaning	Status
A_{geom}	Physical area of the crystal face: 5×5 cm $= 25$ cm^2	Fixed by the design (*Assumed* that the whole face is open to incoming gamma rays)
$\alpha(E)$	Probability that a gamma ray of energy E hitting the face interacts with the crystal	*Derived* from NIST data [12]
$A_{\text{interact}}(E)$	$A_{\text{geom}} \times \alpha(E)$	*Derived*; an upper limit on A_{eff}, not the same thing
$\varepsilon_{\text{optics+SiPM}}(E)$	Fraction of the flash's light that is collected and turned into a pulse (light collection times the SiPM's PDE, and related effects)	**TBD**; to be measured in the lab
$\varepsilon_{\text{thr}}(E)$	Fraction of pulses that are large enough to pass the comparator threshold	**TBD**; to be measured with a threshold scan
$A_{\text{eff}}(E)$	$A_{\text{geom}} \times \alpha \times \varepsilon_{\text{optics+SiPM}} \times \varepsilon_{\text{thr}}$	**TBD** until the efficiencies are measured

The working definition is

$$A_{\text{eff}}(E) \approx A_{\text{geom}} \times \alpha(E) \times \varepsilon_{\text{optics+SiPM}}(E) \times \varepsilon_{\text{thr}}(E).$$

Other missions publish $A_{\text{eff}}(E, \theta)$, where θ is the angle of the incoming gamma ray, from Geant4 or MEGAlib simulations or from in-orbit measurements. Examples are the GRBAlpha maps of effective area and BurstCube's simulated effective area of about 70% of a Fermi-GBM sodium-iodide detector at 100 keV per crystal [4, 7]. PsiSAT currently has only A_{interact}, which is an upper limit on A_{eff}.

7.2 How each factor will be measured or bounded

- A_{geom}: check the open area of the crystal face in the mechanical design (CAD) drawings. The window and housing edges may make it smaller than 25 cm^2.
- $\alpha(E)$: the NIST-based table is used. A Geant4 simulation that includes the window may be used as a cross-check.
- $\varepsilon_{\text{optics+SiPM}}$: measure on the bench with a known light source or radioactive sources. The SiPM data sheet's PDE is used as a first estimate. Different wrapping options will be compared.
- ε_{thr}: change the comparator threshold step by step while the detector is exposed to Am-241 (59.5 keV), Ba-133, and Cs-137 (662 keV) sources. On the bench only, the pulse heights may also be recorded as a diagnostic. This does not change what Path A claims for flight.
- Off-axis gamma rays: the effective area falls with the angle at which the gamma rays arrive and with the longer path through the crystal. Because the orientation is not controlled, an average over orientations will be used (roadmap step 6).

7.3 Energy range

The energy range that PsiSAT Path A can detect has not been established yet. The lowest detectable energy depends on the light collection, the SiPM gain, and the comparator threshold, and will be known after the threshold scan. The highest useful energy is limited by the falling interaction probability $\alpha(E)$ and by how much energy each gamma ray leaves in the crystal. No calibrated energy range will be advertised until it has been measured.

8. Background radiation in orbit

A hard X-ray or gamma-ray detector in low Earth orbit sees a mixture of several signals: the diffuse glow of cosmic X-rays, gamma rays coming from Earth's atmosphere, charged particles trapped in Earth's magnetic field (especially in the SAA and near the poles), secondary particles made by cosmic rays, and radioactivity induced in the satellite itself. It also sees SiPM dark pulses, which increase with radiation damage and with temperature [1, 4]. A light satellite produces less of some kinds of background than a very large spacecraft [1], but the particle rates in low Earth orbit are still large.

8.1 Main sources

Source	Effect on PsiSAT	How other missions deal with it
Diffuse cosmic X-rays and soft gamma rays	A steady background from all directions in the field of view	Included in the rate models [1]
Gamma rays from Earth's atmosphere	Bright when Earth is in the field of view	The orientation and the blocking by Earth must be taken into account [1]
Trapped charged particles in the SAA	Very high count rates, so science data are often discarded	GRBAlpha mapped the region and disabled its trigger there [4]
Regions near the poles (in high-inclination orbits)	Belts of raised background	GRBAlpha's usable time is about 67% [4]
SiPM dark pulses and afterpulses	Raise the lowest detectable energy as the mission goes on	GRBAlpha and GRID track the aging of their SiPMs [1, 4]
Charged particles hitting the crystal	Path A has no way to separate them from gamma rays, which limits the purity of the sample	Admitted as a limit; intervals with high rates are flagged

8.2 What GRBAlpha measured (a reference, not a PsiSAT result)

During commissioning, GRBAlpha mapped its count rate against position over Earth. The SAA and the polar regions were clearly higher. Elsewhere, the background over the full energy range was about 100 counts per second on its larger detector [4] (*Measured* by GRBAlpha, not by PsiSAT).

The fraction of time in which it could detect an average GRB at the 5-sigma level was about 67% on a 550 km polar orbit [4]. PsiSAT's background rates are **TBD**, because its area, thickness, threshold, and orbit are different.

8.3 Operating rule for PsiSAT Path A

Data taken outside the SAA and outside other high-rate periods are treated as valid science data. Without a way to reject charged particles, the purity of the burst sample is limited. Comparing PsiSAT's flagged times, on the ground, with burst announcements from the GCN is an acceptable way to operate a pathfinder. GRBAlpha did the same in its early operations [4]. A full orbit-dependent background model is a deliverable of roadmap step 4 and is **TBD**.

9. Detecting a burst: the trigger and false alarms

In Path A the satellite counts pulses in short time bins of width Δt. The candidate widths are 0.1 s, 1 s, 4 s, and 10 s. The final choice is **TBD**. It also keeps a running estimate of the normal (background) count rate and how much it varies, calculated over a longer stretch of time. Near the SAA the background changes quickly, so the rules for freezing or updating this estimate there still need to be set. Other missions work in a similar way: Fermi-GBM looks for rate rises on several time scales, and GRBAlpha used continuous binned counts (4 s bins at first, 1 s bins later) and studied when to turn its trigger on or off [4, 6].

9.1 Draft trigger rule (to be completed in roadmap step 3)

The satellite raises a trigger if the counts above the expected background, N_{ex}, in a bin of width Δt are at least k times the expected random scatter of the background, σ_{bg}, for M bins in a row:

$$N_{\mathrm{ex}} \geq k \cdot \sigma_{\mathrm{bg}}.$$

The starting value is $k \sim 5$, until a proper calculation of the false-alarm rate replaces it. For each trigger the satellite will record the mission elapsed time, the counts in each bin, the background estimate μ_b and its scatter σ_{bg}, and flags: whether it was in the SAA, whether the counter was saturated, and whether the trigger had been switched off by command. *Dead time* is the short time after each event during which the electronics cannot record another. If pulses arrive too fast, the counter saturates. We will measure the maximum pulse rate the STM32 input can handle on the bench and define how the satellite behaves when it is saturated.

9.2 False alarms

Without energy measurements (no PHA), it is harder to tell real bursts from particle showers and noise spikes. Planned measures are: (1) discarding data in the SAA and during other high-rate periods; (2) requiring the rise to show up on several time scales; (3) comparing on the ground with the GCN and with the Fermi and Swift observatories; and (4) adding pulse-height analysis in a later Path B. The expected false-alarm rate is **TBD**, until the background model and the trigger algorithm are complete. Flags produced on board will not be presented as a catalog of GRBs until their purity has been shown.

10. Timing and localization

10.1 What the Path A timing provides

The STM32 records the time of each pulse and each trigger relative to the start of the mission (mission elapsed time). This is enough for building light curves (plots of counts against time), for looking at the data around a burst time announced by another observatory, and for teaching how to operate a trigger.

10.2 What the Path A timing does not provide

Without a GPS receiver and its precise timing signal, the timestamps are not tied to universal time. They cannot be used to match bursts with LIGO gravitational-wave detections or to find a burst's direction from arrival-time differences between satellites at the accuracy that astronomy needs. The absolute timing accuracy is **TBD**, and is not needed until Path C. The drift of the satellite's clock will still be measured on the bench.

10.3 Why one satellite cannot locate a burst

A single flat detector on a satellite that tumbles or is slowed by a magnet cannot find the direction of a GRB. Even rough locations need several detectors pointing in different directions (as on BurstCube) or several satellites with accurate timing (as planned for CAMELOT and HERMES) [1, 4, 7]. Path A only claims to detect count-rate rises and record their times.

11. Calibration and test plan

11.1 Minimum bench tests

Test	What we will do	What it tells us
Dark rate versus bias voltage and temperature	Measure the SiPM alone, then with the crystal, while changing the temperature in a chamber	The noise floor and a baseline for later aging
Threshold scan	Change the comparator level step by step and count pulses from radioactive sources	$\varepsilon_{\mathrm{thr}}$, and how to convert the threshold into keV
Radioactive sources	Use Am-241 (59.5 keV), Ba-133, and Cs-137 (662 keV)	Reference points for the response; optionally a lab pulse-height analyzer (MCA) that sorts pulses by height
Dead time	Send pulses at a high rate into the STM32 input	How the counter saturates
Optical options	Compare with and without the reflective film (ESR) and optical gel, and the quality of the coupling	A bound on $\varepsilon_{\mathrm{optics+SiPM}}$
Temperature drift	Measure gain and threshold against temperature	Safe operating margins and how to set the SiPM bias voltage
Comparison with a lab detector	Measure side by side with a laboratory PMT or SiPM setup, if available	A sanity check

As an optional extra, a simple pulse-height measurement may be made on the bench, only as a diagnostic. It does not permit any claims about soft or hard bursts in flight (that would be Path B).

11.2 Calibration in flight

Path A carries no radioactive calibration source. In flight we will track the count rate against temperature, change the threshold periodically by command, compare the flagged times with public GCN burst times, and follow the long-term rise of the SiPM's dark rate, as GRBAlpha and GRID did [1, 4]. A proper energy calibration in flight needs Path B.

12. Performance numbers and their status

Each number below is labeled as *Measured*, *Derived*, *Assumed*, or **TBD** (see Section 2).

Quantity	Value	Label	Source or note
Crystal face	50×50 mm	*Assumed*	From the parts list and design
Thickness x	3.0 mm	*Assumed*	From the design
A_{geom} (head-on)	25 cm^2	*Assumed*	5×5 cm; the open area must still be confirmed
ρ (CsI)	4.51 g/cm^3	*Assumed*	Typical for crystalline CsI; a change of ± 0.02 changes α by less than about 1%
μ/ρ (100 keV)	2.035 cm^2/g	*Measured**	NIST XCOM table for CsI [12] (*from a published table)
μ (100 keV)	9.18 cm^{-1}	*Derived*	$(\mu/\rho) \cdot \rho$
α (100 keV)	93.6%	*Derived*	$1 - e^{-\mu x}$; interaction probability
A_{interact} (100 keV)	23.4 cm^2	*Derived*	$A_{\text{geom}} \cdot \alpha$; upper limit on A_{eff}
α (200 keV)	40.2%	*Derived*	From the NIST-based table
α (300 keV)	21.8%	*Derived*	From the NIST-based table
$\varepsilon_{\text{optics+SiPM}}$	—	**TBD**	To be measured in the lab
ε_{thr}	—	**TBD**	To be measured with a threshold scan
$A_{\text{eff}}(E)$	—	**TBD**	Needs the two efficiencies above
Energy range in flight	—	**TBD**	Will be set by the threshold scan
Background rate in orbit	—	**TBD**	Depends on the orbit and the threshold
Fraction of time usable	—	**TBD**	GRBAlpha's ∼67% is for that mission, not for PsiSAT [4]
GRBs detected per year	—	**TBD**	A method only, until A_{eff} is known
Faintest detectable burst (fluence)	—	**TBD**	Needs A_{eff} and the background
Absolute timing	Mission elapsed time	*Assumed*	No GPS in Path A
Localization	None	*Assumed*	One flat detector

Table 3: Performance numbers with their labels.

12.1 How a rate estimate will be made

A clear estimate of the number of bursts detected per year would take the form

$$\dot{N} \approx \int R(F) \cdot P(\text{detect} \mid F, A_{\text{eff}}, \Delta t, \text{bg}, \text{duty}, \text{FOV})\, dF,$$

where $R(F)$ is the rate of bursts of fluence F (the total energy per unit area that a burst delivers), and P is the probability of detecting such a burst given the effective area, the bin width, the background, the fraction of time observing (duty), and the field of view (FOV). The burst rates and brightness distributions would come from the Fermi-GBM catalog [6], with all assumptions stated. PsiSAT does not publish a number of GRBs per year in this document, because the effective area, threshold, background, and fraction of usable time are still **TBD**. If a number is added in a later revision, every assumption behind it will be listed next to it.

13. Risks and limitations

Risk or limitation	Effect	What we will do
Effective area unknown until lab tests are done	Sensitivity and detection rates cannot yet be defended	Carry out the lab plan (roadmap step 7); simulate later with Geant4
No energy measurement (no PHA), so charged particles are hard to reject	False triggers and an impure sample	Discard SAA data; compare with GCN announcements; add PHA in Path B
High background in the SAA and near the poles	Loss of usable observing time	Flag these periods in the data
Radiation damage to the SiPM	The lowest detectable energy rises over months to years	Keep margin in the bias voltage; monitor the count rate [1, 4]
Temperature changes the SiPM gain	The threshold drifts	Use a temperature sensor; test at different temperatures
No GPS	No gravitational-wave matching; no arrival-time localization	Left out of Path A; planned for Path C
Uncontrolled orientation	The exposure is unknown and changes with time	Use an average-exposure model; make no claim about pointing away from Earth
Single detector	Cannot locate bursts	Stated as a limitation
Student schedule and the CSLI timeline	Some parts of the proposal may be unfinished	Make each physics claim only after the matching roadmap step is done
Faculty principal investigator (PI) not yet confirmed	Faculty endorsement is still open	Arrange a meeting and the paperwork with Prof. Will Farr

14. Roadmap for the physics work

Hardware work can go on in parallel. A physics claim is made only when the matching step is finished.

Step	Name	Status (Sep 2026)	Result
0	Fix the scope	Done (draft)	A list of what is and is not claimed, and a one-sentence goal
1	Detection principle	Done (draft)	The signal chain and the interaction physics
2	Instrument response	Started; largely done in this document	Crystal area and NIST-based table; efficiencies **TBD**
3	Trigger specification	Not started	A trigger algorithm based on a calculated false-alarm rate
4	Background in orbit	Not started (Section 8 is a start)	SAA, false alarms, fraction of usable time
5	Sensitivity and rates	Method to do; number **TBD**	A rate estimate that can be defended, or keep it **TBD**
6	Orientation and exposure	Not started	An average-exposure model for a randomly oriented satellite
7	Lab plan	Not started (Section 11 is a start)	Dark rate, threshold scan, dead time
8	Physics chapter of the proposal	This document	The Path A physics chapter for CSLI
9	Upgrades: Path B and Path C	Later	Energy measurement and GPS time, only after they are built and tested

The physics work for the first version is complete when: the scope is fixed; the detection and response calculations are written; the trigger can be implemented; a background model exists; the method for estimating sensitivity is defined; the lab plan is written; and this physics chapter is finished.

15. Summary

PsiSAT Path A is an educational 1U CubeSat with a CsI(Tl) crystal and SiPM that counts gamma rays and records when they arrive, to observe gamma-ray bursts in low Earth orbit. It is proposed to NASA CSLI by Muhammad Maalik, Jackson Boyle, and Christian Tumanda at Stony Brook University, with Prof. Will Farr as the intended faculty PI (confirmation pending).

The crystal calculation is done: from NIST data, $\alpha(100\,keV) \approx 93.6\%$ for 3 mm of CsI. This is the fraction of gamma rays hitting the face that interact with the crystal. It is a necessary step towards the effective area, but far from enough by itself. The effective area, the trigger algorithm, the background and usable fraction of time, the lab calibration, the GPS timing, and the detection rate are all still open, and each has a plan to be closed. Similar missions, especially GRBAlpha, show that small CsI-and-SiPM CubeSats can detect real GRBs when their claims match their hardware. PsiSAT is one or two steps behind that mission, which also measures energy, and this proposal states plainly what remains to be done.

This proposal claims only what the planned hardware can support.

16. References

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Appendix A. NIST-based interaction table

The table assumes a head-on beam, $\rho = 4.51$ g/cm^3 (*Assumed*), $x = 0.3$ cm, and $A_{\text{geom}} = 25$ cm^2. The mass attenuation coefficient μ/ρ is from NIST XCOM for CsI [12]. The other columns are calculated (*Derived*): $\mu = (\mu/\rho) \cdot \rho$, $\alpha = 1 - e^{-\mu x}$, and $A_{\text{interact}} = A_{\text{geom}} \cdot \alpha$. The effective area A_{eff} is **TBD**.

E (keV)	μ/ρ (cm^2/g)	μ (cm^{-1})	α	A_{interact} (cm^2)	A_{eff}
20	26.86	121.140	1.0000	25.00	TBD
30	9.045	40.793	1.0000	25.00	TBD
33.17	6.923	31.223	0.9999	25.00	TBD
40	22.97	103.590	1.0000	25.00	TBD
50	12.87	58.044	1.0000	25.00	TBD
60	7.921	35.724	1.0000	25.00	TBD
80	3.677	16.583	0.9931	24.83	TBD
100	2.035	9.178	0.9363	23.41	TBD
150	0.729	3.288	0.6271	15.68	TBD
200	0.3805	1.716	0.4024	10.06	TBD
300	0.1818	0.820	0.2181	5.45	TBD
400	0.1237	0.558	0.1541	3.85	TBD
500	0.09809	0.442	0.1243	3.11	TBD
600	0.08373	0.378	0.1071	2.68	TBD
800	0.06769	0.305	0.0875	2.19	TBD
1000	0.05848	0.264	0.0760	1.90	TBD

Table 4: Table A1. Near 33–40 keV, the absorption of iodine and cesium jumps sharply (the "K-edge"), so μ/ρ changes suddenly there. The transmission of the crystal window is not included.

Appendix B. Open items

ID	Open item	How it will be closed
T1	Effective area $A_{\mathrm{eff}}(E) = A_{\mathrm{geom}} \cdot \alpha \cdot \varepsilon_{\mathrm{optics}} \cdot \varepsilon_{\mathrm{thr}}$	Lab threshold and optics measurements; optional Geant4 simulation
T2	Trigger algorithm (bin width Δt, k, M, SAA rule)	Write a specification in roadmap step 3 that a student programmer can implement
T3	Background model for the orbit, including the SAA	Roadmap step 4; use maps published by other missions as first limits
T4	Lab calibration (Am-241, Ba-133, Cs-137, and the effect of temperature)	Roadmap step 7 test campaign
T5	GPS and absolute timing accuracy	Only with Path C; until then, mission elapsed time is used
T6	Expected number of GRBs detected per year	Only after T1 to T3, using the Fermi-GBM catalog [6]
T7	Open area of the crystal face compared with 25 cm^2	Measure from the CAD drawings of the window and housing
T8	Lowest detectable energy in flight, in keV	From the threshold scan in T4
T9	False-alarm rate	From T2 and T3
T10	Faculty PI's approval of the scope	Meeting with Prof. Will Farr

End of PsiSAT Physics Proposal: Path A — 24 September 2026. This document is the physics chapter of the PsiSAT proposal.